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R PROGRAMMING PROJECT

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1. Introduction

R is a powerful programming language and software environment used for statistical computing, data analysis, and graphical visualization. It is widely used by data scientists, statisticians, and researchers.

R was developed by Ross Ihaka and Robert Gentleman. It is open-source and provides a wide range of tools for data manipulation, calculation, and graphical display.

It supports various statistical techniques such as:

- Regression analysis
- Classification
- Clustering
- Time-series analysis

R is one of the most popular tools in data science due to its flexibility and strong community support.

2. Objectives

- To understand the basics of R programming
- To perform statistical calculations
- To create and analyze datasets
- To visualize data using graphs
- To interpret results effectively

3. Tools & Requirements

- R Software
- RStudio IDE (optional)
- Computer system
- Basic programming knowledge

4. Dataset Description

We are using a dataset of student marks:

Student	Marks
S1	70
S2	85
S3	90
S4	60
S5	75

This dataset is used to perform statistical analysis.

5. R Program Code

```
# Create dataset
marks <- c(70, 85, 90, 60, 75)

# Display data
print("Marks:")
print(marks)

# Statistical calculations
mean_marks <- mean(marks)
median_marks <- median(marks)
sd_marks <- sd(marks)

max_marks <- max(marks)
min_marks <- min(marks)

print(mean_marks)
print(median_marks)
print(sd_marks)
print(max_marks)
```

```
print(min_marks)
```

```
# Bar Plot
```

```
barplot(marks,  
        col="blue",  
        main="Student Marks",  
        xlab="Students",  
        ylab="Marks")
```

```
# Pie Chart
```

```
pie(marks,  
    main="Marks Distribution",  
    col=rainbow(length(marks)))
```

6. Explanation of Code

- `c()` creates a vector
- `mean()` calculates average
- `median()` finds middle value
- `sd()` calculates variation
- `max()` and `min()` find highest & lowest values
- `barplot()` creates bar chart
- `pie()` creates pie chart

7. Output

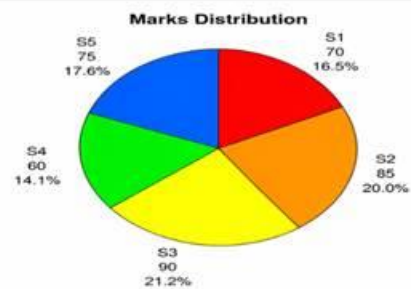
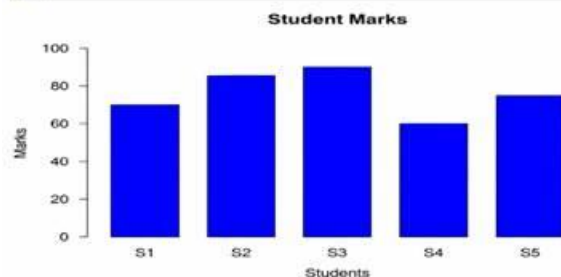
- Mean = 76
- Median = 75
- Standard Deviation ≈ 11.4
- Maximum = 90
- Minimum = 60

The program successfully performs calculations and generates graphs.

8. Graphical Representation

- **Bar Chart:** Shows comparison of marks
- **Pie Chart:** Shows distribution of marks

```
> # Create dataset
> marks <- c(70, 85, 90, 60, 75)
>
> # Display data
> print("Marks:")
[1] "Marks:"
> print(marks)
[1] 70 85 90 60 75
>
> # Statistical calculations
> mean_marks <- mean(marks)
> median_marks <- median(marks)
> sd_marks <- sd(marks)
>
> max_marks <- max(marks)
> min_marks <- min(marks)
>
> print(mean_marks)
[1] 76
> print(median_marks)
[1] 75
> print(sd_marks)
[1] 11.40175
> print(max_marks)
[1] 90
> print(min_marks)
[1] 60
>
```



9. Applications of R Programming

- Data Science & Analytics
- Machine Learning

- Financial Analysis
- Healthcare & Bioinformatics
- Business Intelligence

10. Advantages

- Free and open-source
- Strong statistical capabilities
- Excellent visualization tools
- Large community support
- Wide range of packages

11. Disadvantages

- Slower performance in some cases
- High memory usage
- Not ideal for large-scale applications
- Learning curve for beginners

12. Conclusion

R programming is a powerful tool for statistical analysis and data visualization. This project demonstrates how R can be used to analyze data, perform calculations, and generate meaningful insights.

13. Future Scope

- Use real-world datasets
- Apply machine learning models
- Integrate with Python
- Build advanced analytics systems

